You are requested to write a paper of 3000 words i.e. 6 pages, about one of the listed below cloud computing research topics in which you will be researching all aspects of the chosen topic.

In your research you will need to define and analyze clearly the topic under research. Discuss all information related to the topic showing all techniques, technologies used and tools if applicable. Using illustrations and images in your paper is highly recommended to provide a better understanding of the topic. At the end of your paper you will need to provide a conclusion paragraph for your research according to your critique\opinion about the topic and the research findings. A references section at the end of the paper is required and needs to be alphabetically sorted.

Guidelines:

- The paper should be between 3,000 – 3,500 words i.e. 6 pages.
- Font used for body would be Times New Roman 12 points.
- Font used for Titles would be Times New Roman 12 point in Bold.
- Font used for Image\illustration descriptions would be Times New Roman 10 points in Italics.
- The deliverable is a Microsoft word file showing the name and ID of the student.
- A cover page at the beginning showing the research topic, name and ID of the student is required.
- References Section at the end of the paper should be sorted alphabetically.
- Plagiarism over 15% will result in losing marks.
- The use of AI is totally prohibited resulting in 0 Grade.
- Due date is Thursday, November the 20th, 2025 at 9.00 p.m. by submission on Moodle platform.

Research Topics:

1. Green Cloud Computing The major challenge in the cloud is the utilization of energy-efficient and hence develop economically friendly cloud computing solutions. Green cloud computing provides solutions to make resources more energy efficient and to reduce operational costs. This pivots on power management, virtualization of servers and data centers, recycling vast e-waste, and environmental sustainability.

2. Edge Computing It is the advancement and a much more efficient form of Cloud computing with the idea that the data is processed nearer to the source. Edge Computing states that all of the computation will be carried out at the edge of the network itself rather than on a centrally managed platform or the data warehouses. Edge computing distributes various data processing techniques and mechanisms across different positions. This makes the data deliverable to the nearest node and the processing at the edge. This also increases the security of the data since it is closer to the source and eliminates late response time and latency without affecting productivity.

3. Mobile Cloud Computing In mobile cloud computing, the mobile is the console and storage and processing of the data takes outside of it. It is one of the leading Cloud Computing research topics. The main advantage of Mobile Cloud Computing is that there is no costly hardware and it comes with extended battery life. The only disadvantage is that has low bandwidth and heterogeneity.

4. Cloud Cryptography Cloud cryptography is the practice of securing data and communications in cloud computing environments using cryptographic methods and protocols. Sensitive data is secured against unauthorized access and possible security breaches by encrypting it both in transit and at rest. By allowing consumers to keep control of their data while entrusting it to cloud service providers, cloud cryptography protects the confidentiality, integrity, and authenticity of that data. Cloud cryptography improves the security posture of cloud-based apps and services, promoting trust and compliance with data privacy rules by using encryption methods and key management procedures.

5. Big Data Big data is the technology denotes the tremendous amount of data. This data is classified in 2 forms that are structured (organized data) and unstructured (unorganized). This can be used for research purpose and companies can use it to detect failures, costs, and issues. Big data is characterized by three Vs which are:

- Volume – It refers to the amount of data which handled by technologies such as Hadoop.
- Variety – It refers to the present format of data.
- Velocity – It means the speed of data (generation and transmission).

6. Cloud Security Since the number of companies and organizations using cloud computing is increasing at a rapid rate, the security of the cloud is a major concern. Cloud computing security detects and addresses every physical and logical security issue that comes across all the varied service models of code, platform, and infrastructure. It collectively addresses these services, however, these services are delivered in units, that is, the public, private, or hybrid delivery model. Security in the cloud protects the data from any leakage or outflow, theft, calamity, and removal. With the help of tokenization, Virtual Private Networks, and firewalls data can be secured.

7. DevOps DevOps is an amalgamation of two terms, Development and Operations. It has led to Continuous Delivery, Integration, and Deployment and therefore reducing boundaries between the development team and the operations team. Heavy applications and software need elaborate and complex tech stacks that demand extensive labor to develop and configure which can easily be eliminated by cloud computing. It offers a wide range of tools and technologies to build, test, and deploy applications with a few minutes and a single click. They can be customized as per the client requirements and can be discarded when not in use hence making the process seamless and cost-efficient for development teams.

8. Cloud Load Balancing It refers to splitting and distributing the incoming load to the server from various sources. It permits companies and organizations to govern and supervise workload demands or application demands by redistributing, reallocating, and administering resources between different computers, networks, or servers. Cloud load balancing encompasses holding the circulation of traffic and demands that exist over the Internet. This reduces the problem of sudden outages, results in an improvement in overall performance, has rare chances of server crashes, and also provides an advanced level of security. Cloud-based servers' farms can accomplish more precise scalability and accessibility using the server load balancing mechanism. Due to this, the workload demands can be easily distributed and controlled.